Project - Valuation and Hedging - Options

1. Select a stock in order to create an option. Download the daily market data (use the adjusted closed price as it exclude dividends).

2. Choose and estimate the necessary parameters for the pricing of an option:

   a) Select an annualized risk-free rate “r” - a Treasury Bill, European Bond, etc. Precise your source.

   b) Choose a projection period “T” (“T” should be chosen for the same period as “r”).

   c) Calculate the annualized volatility - a common rule of thumb is to use data covering a period equal to the length of “T”.

3. Black-Scholes approach:

   a) Define the Black-Scholes model. Explain its assumptions.

   b) Using the Black-Scholes formula, estimate the price of a call option using as strike price the mean “μ” of your asset’s market price data.

   c) Estimate the price of the respective put using the parity condition. Estimate the payoff and the PnL of the put option.

4. Define the Greeks and explain their characteristics. For the call option in 3 b), calculate Delta, Gamma, Theta (daily), Vega (assume 2.5% increase in volatility), Rho (assume 1.4% increase in the risk-free rate). Interpret each of your results.

5. Define the notion of implied volatility. Explain its characteristics. Calculate the implied volatility of the call option in 3 b) and of the put option in 3 c). To define the market price of your options, you can either search online for options with the same underlying as a reference, or assume a 25% deviation (positive or negative) from the market price data for your asset.

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6. Stochastic processes:
   a) Define the Geometric Brownian Motion process.
   b) Using the Monte Carlo approach, estimate the price of a call option using as strike price "K" the mean "μ" of your asset's market price data. To price the underlying, use the GBM process. Perform 20 000 simulations.
   c) Define the Local Volatility process (optional).
   d) Using the Monte Carlo approach, estimate the price of a call option using as strike price "K" the mean "μ" of your asset's market price data. To price the underlying, use the Local Volatility process. Perform 20 000 simulations. (optional)
      - To simply the calculation, assume that $\mu = (r - q)$.
      - To approximate the partial derivatives using finite differences, make logical assumptions regarding the strike prices and maturities you need in order to estimate the options $V_{K,T+1}$ and $V_{K,T-1}$, and the options $V_{K+1,T}$ and $V_{K-1,T}$.

7. Define hedging. What is hedging? Why use hedging strategies? Who uses hedging strategies? What are the main approaches?

8. Delta hedging (optional)
   a) Assuming you are long 25 000 shares of your chosen asset, for the call option in 3 b), determine the number of options necessary to create a delta-neutral portfolio (use a Covered Call or a Protective Put strategy). Interpret the result.
   b) Estimate the effect on your delta-neutral portfolio if the market price of your asset increases by $1.5. Interpret the result.
   c) What is the impact on delta after the $1.5 increase? How should you rebalance your portfolio to maintain the delta-hedge? Make the necessary calculations and interpret the result.

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9. Gamma hedging (optional)

a) Assuming the same investment portfolio as in 8 a), create a gamma-neutral position in order to protect your portfolio from market risk. To hedge, select a random option (specify your source) and its respective delta and gamma values.

b) Is it possible to adjust your portfolio to maintain a delta-gamma neutral position ? Make the necessary calculations and interpret the result.

10. Vega hedging (optional)

a) Assume you have an existing call spread position consisting of the long call option from 3b) and a short call option, with the same parameters as the option from 3b) except that the strike price is 30% higher than the strike price of the option from 3b). How can you adjust your portfolio to achieve a vega-neutral position? Perform the necessary calculations and interpret the result.

11. Theta hedging: Assuming you have an existing butterfly position, how can you adjust your portfolio to achieve a theta-neutral position? Choose the options you would need to create such a portfolio, perform the necessary calculations and interpret the result. (optional)

12. Estimate the price of an American call option using the Binomial Options Pricing Model detailed in Annex 3. (new question, mandatory)

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Requirements:

- Groups of 2 people
- Deliverables: Dataset + Python script
- Deadline: Sunday - 8th of March: 21h

<u>Note</u>: Do not use a Python library for the Black-Scholes, implied volatility, GBM and Local Volatility models, write their equations as shown in the slides.